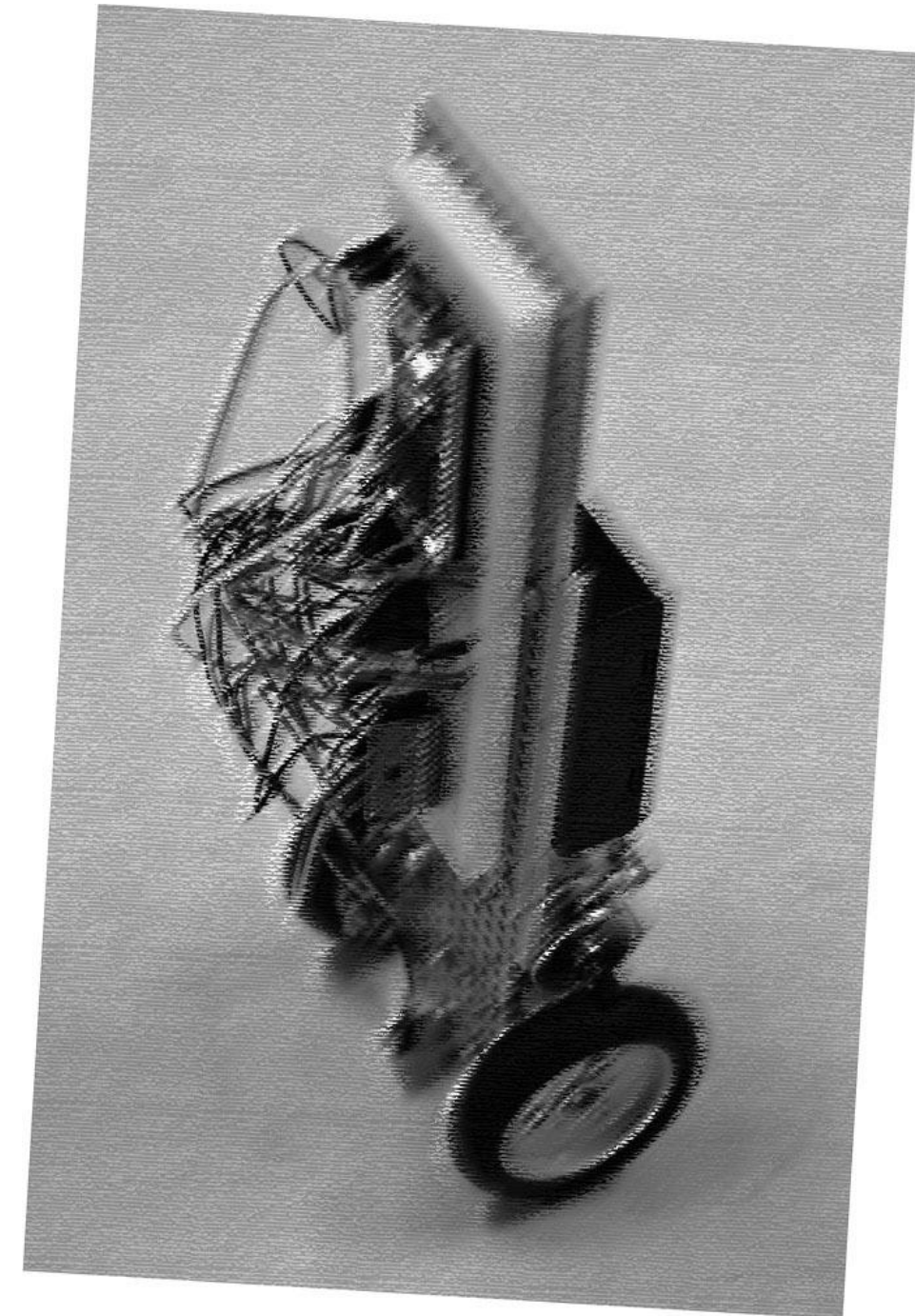


PRACTICAL EXERCISE · WEEK 4

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE Department of Precision Engineering · 2026 S1S2
SESSIONS Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14



Timeline

1

Week 1 Standing

Assembly & bring-up
Angle-only **standing** demo

2

Week 2 Position control

Build a rotary encoder
Position control
Simulation in Simulink

3

Week 3 EXTENSION

Load & disturbance tests
Advanced control (LQR/MPC)
.....

4

Week 4 presentation

10-minute presentation
Live demo & Q&A

WEEK 4 · FINAL SHOWCASE PREVIEW

8 slides - 7 minutes + 3-minute demo + 5-min Q&A

01 Introduction Project goal, team task, robot configuration. State the problem and why it matters.	02 Method Modeling, control design, sensor usage, implementation strategy.	03 Experiment Test setup, conditions, evaluation procedure. How did you measure success?	04 Data Key plots, tables, numerical results — the evidence behind your verdict.	05 Extended Experiment Beyond the basic task — load test, parameter study, sim vs. real, alternative controller.
06 Results & Future Work What worked, what didn't, what to improve next. Honest assessment over hype.	07 What You Learned Reflect on lessons — technical, experimental, teamwork, debugging.	08 Teamwork & Roles Who did what — task split, collaboration, individual contributions and shared decisions.	+3 MINUTES Demo Video or Live Run A video or run the robot live	

Week 4 Submission

Slides (.pptx)

Microcontroller Code (.cpp)

Final Demo Video